



KCuD - 400

Copper-diamond material for thermal

KEY PROPERTIES

- Isotropic material
- Extremely high thermal conductivity
- Tailored CTE (matching capability)
- High thermal diffusivity
- Wide range of geometries available

APPLICATIONS

- Diode laser systems
- Electrical drives
- Wireless communications
- Heat dissipation of GaN devices

MAIN PROPERTIES

Property	Value	Unit
Density	5.30	g/cm ³
Flexural strength	195	MPa
Electrical Conductivity	9	MS/m
Thermal Conductivity	432	W/m·K
CTE	8.6	x10 ⁻⁶ K ⁻¹
Specific Heat	0.41	J/g·K
Thermal Diffusivity	199	mm ² /s

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)		600 °C
Recomended temp		<500 °C
Continuous temp		450 °C

ELECTRICAL BEHAVIOR

Conductive	
Electrical Conductivity	9MS/m

MATERIAL INFORMATION

Material type		Copper-diamond composite
Structure		Metal matrix composite
Key feature		Thermal management
Machinability		Medium
Color		Metallic

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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